

Freeze ablation – results

Three runs sharing the same setup (HCP-only, T=1200, temporal_kernel=20, base_lr=3e-4, effective batch 16, 20 epochs, DINOv2 ViT-S/14 reg4 ImageNet init). Only the parameter-freezing policy differs.

Strategies

Strategy	Trained components	Trainable params
A – Full fine-tune	full backbone + heads	~22 M
B – Freeze last3	patch_embed.* + blocks.9–11 + norm + heads	~3.6 M
C – Freeze except input	patch_embed.* + heads	~300 k

Best checkpoint per strategy (selected among available iters: 5999/8999 for A, 11999/17999 for B and C).

HCP probes

Label	Metric	A (full FT)	B (freeze last3)	C (freeze fmri)
Sex	AUC	0.751	0.696	0.844
BrainVol	MAE	116,179	117,744	99,455
GrayMatterVol	MAE	34,762	34,815	29,274
Age	MAE	3.044	3.056	2.931
FluidIntel	MAE	4.047	4.035	4.003
ProcSpeed	MAE	16.343	16.401	16.207
WorkingMem	MAE	10.442	10.410	10.411

ADNI probes

Label	Metric	A (full FT)	B (freeze last3)	C (freeze fmri)
Sex	AUC	0.653	0.696	0.813
CDR	AUC	0.542	0.522	0.532
Degradation1Y	AUC	0.581	0.543	0.559
Degradation2Y	AUC	0.533	0.456	0.533
Degradation3Y	AUC	0.528	0.468	0.544
MMSE	MAE	2.478	2.426	2.750
Age	MAE	5.113	4.993	5.703

Bold = best value in row. AUC: higher is better. MAE: lower is better.

Reproducibility

- Repo / branch: <https://github.com/DanAbergel/dinov2>, branch fmri-mixed-t140.
- YAML configs: dinov2/configs/train/fmri_vits.yaml (A), fmri_vits_hcp_freeze_last3.yaml (B), fmri_vits_hcp_freeze_fmri.yaml (C).

- Single sbatch: `slurm_jobs/run_dinov2_fmri.sh`, switch via `CONFIG_FILE=...`.
- Probes: `slurm_jobs/probe_adni.sh`, `slurm_jobs/probe_hcp.sh` (auto-launched by `slurm_jobs/probe_3ckpts.sh`).
- Aggregation: `python3 scripts/summarize_probes.py`.